Year 12 Further Maths - Momentum and collisions

Similar questions - Walls, strings and repeated collisions

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Particles A and B, of masses 3 kg and 5 kg, are connected by a light inextensible string and are at rest on a smooth horizontal surface with the string slack. A is projected directly away from B with a speed of $8\ \mathrm{m\,s^{-1}}$.

(a) Find the speed of the particles immediately after the string becomes taut. **[3]**

(b) Find the magnitude of the impulse exerted on B by the string. **[2]**

(c) Find the loss in kinetic energy when the string becomes taut. **[3]**

(d) If A has mass m and B has mass $2m$ and A is projected with speed u, find in terms of m and u the impulse on B. **[2]**

2 A particle P of mass 3 kg is moving with speed $6\ \mathrm{m\,s^{-1}}$ on a smooth horizontal plane when it strikes a fixed vertical wall at right angles. It rebounds with speed $4.2\ \mathrm{m\,s^{-1}}$.

(a) Find the coefficient of restitution between P and the wall. **[1]**

(b) Find the magnitude and state the direction of the impulse exerted on P by the wall. **[3]**

(c) State the magnitude and direction of the impulse exerted on the wall by P. **[2]**

P now travels between two parallel fixed smooth walls, striking them alternately at right angles. The coefficient of restitution is the same for each wall.

(d) Find the number of rebounds after which the speed of P is first less than $1\ \mathrm{m\,s^{-1}}$. **[4]**

3 Particles A of mass $3m$ and B of mass m are on a smooth horizontal floor. A is moving with speed u directly towards a vertical wall, and B is at rest between A and the wall. A collides directly with B; the coefficient of restitution between them is $\frac{1}{3}$.

(a) Show that, after the collision, the speed of A is $\frac{2}{3}u$ and the speed of B is u. **[5]**

B then collides with the wall, and the coefficient of restitution between B and the wall is f. B then collides with A for a second time. The first collision between A and B occurs at a distance d from the wall. The second collision between A and B occurs at a distance $\frac{1}{7}d$ from the wall.

(b) Find f. **[6]**

(c) Show that, after the second collision between A and B, B will hit the wall again. **[5]**

4 Two particles P and Q are on a smooth horizontal floor. P has mass 2 kg and is moving with speed $3\ \mathrm{m\,s^{-1}}$ directly towards a vertical wall. Q has mass 5 kg and is between P and the wall, moving directly towards P with speed $0.5\ \mathrm{m\,s^{-1}}$. P and Q collide directly and the coefficient of restitution for the collision is e.

(a) Show that, after the collision, the speed of Q towards the wall is $(0.5+e)\ \mathrm{m\,s^{-1}}$. **[5]**

Q then collides directly with the wall. The coefficient of restitution between Q and the wall is also e. There is no subsequent collision between P and Q.

(b) Determine the range of possible values of e. **[7]**

Question 1
(a) Momentum: $3(8) = 8v$, so $v = 3\ \text{m s}^{-1}$
(b) Impulse on $B = 5(3) = 15\ \text{N s}$
(c) Before: $\frac{1}{2}(3)(64) = 96$ J. After: $\frac{1}{2}(8)(9) = 36$ J. Loss $= 60$ J
(d) $mu = 3mv$, so $v = \frac{1}{3}u$ and the impulse on B is $2m \cdot \frac{1}{3}u = \frac{2}{3}mu$

Question 2
(a) $e = 4.2/6 = 0.7$
(b) $I = 3(6+4.2) = 30.6\ \text{N s}$, directed away from the wall
(c) 30.6 N s, directed towards the wall (Newton's third law)
(d) After n rebounds the speed is $6(0.7)^n$. $6(0.7)^5 = 1.008 > 1$ but $6(0.7)^6 = 0.706 < 1$, so after 6 rebounds

Question 3
(a) Momentum: $3mu = 3mv_A + mv_B$. Restitution: $v_B - v_A = \frac{1}{3}u$. So $3u = 3v_A + v_A + \frac{1}{3}u$, $v_A = \frac{2}{3}u$ and $v_B = u$
(b) B reaches the wall after time d/u, when A has moved $\frac{2}{3}d$ and is $\frac{1}{3}d$ from the wall. B then rebounds with speed fu. They meet after a further time t where $fut + \frac{2}{3}ut = \frac{1}{3}d$, so $t = \dfrac{d}{3u(f+\frac{2}{3})}$. Distance from the wall at the second collision $= fut = \dfrac{fd}{3f+2}$. Setting $\dfrac{f}{3f+2} = \dfrac{1}{7}$ gives $7f = 3f+2$ and $f = \frac{1}{2}$
(c) Take the direction towards the wall as positive. Before the second collision: A has velocity $\frac{2}{3}u$, B has velocity $-\frac{1}{2}u$. Momentum: $3m \cdot \frac{2}{3}u - \frac{1}{2}mu = \frac{3}{2}mu = 3mv_A + mv_B$. Restitution: $v_B - v_A = \frac{1}{3}\left(\frac{2}{3}u + \frac{1}{2}u\right) = \frac{7}{18}u$. So $4v_A = \frac{3}{2}u - \frac{7}{18}u = \frac{10}{9}u$, giving $v_A = \frac{5}{18}u$ and $v_B = \frac{2}{3}u$. Both move towards the wall, so B hits the wall again (and B is faster than A)

Question 4
(a) Positive towards the wall. Momentum: $2(3) + 5(-0.5) = 3.5 = 2v_P + 5v_Q$. Restitution: $v_Q - v_P = 3.5e$. So $3.5 = 2v_Q - 7e + 5v_Q$, giving $v_Q = 0.5 + e$ and $v_P = 0.5 - 2.5e$
(b) Q rebounds from the wall with velocity $-e(0.5+e)$. There is no further collision if P is not catching Q: $v_P \leqslant -e(0.5+e)$. So $0.5 - 2.5e \leqslant -0.5e - e^2$, i.e. $e^2 - 2e + 0.5 \leqslant 0$, giving $1 - \frac{1}{\sqrt{2}} \leqslant e \leqslant 1 + \frac{1}{\sqrt{2}}$. With $0 \leqslant e \leqslant 1$: $0.293 \leqslant e \leqslant 1$ (strictly $e > 0.293$ if they are not to touch again)